

University of Sheffield

School of Information, Journalism and Communication

Dissertation Proposal

IJC470: Research Methods and Dissertation Preparation

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1. Introduction

1.1 Background for the Work

Endometriosis is a chronic health condition that affects approximately 190 million females of reproductive age worldwide (WHO, 2023). The prevalence is often considered to be underestimated because the current gold standard for diagnosis involves laparoscopy, an invasive surgical procedure that is not a routine examination (Zondervan et al., 2020; Shafrir et al., 2018). Another, non-invasive, approach to help diagnose endometriosis is pelvic MRI, which offers a safer method, though MRI can achieve over 90% diagnostic sensitivity and specificity only in optimal conditions (Bazot et al., 2004; Deslandes et al., 2020). The diagnostic value of MRI is, however, contingent on the accurate segmentation of the anatomical structures of interest. The issue with MRI is the task of segmenting the possible areas of interest: the uterus, ovaries, and endometriomas (cysts made from endometrial cells). This challenge is twofold: firstly, the affected areas can be quite small and difficult to properly delineate; and secondly, endometriosis can deform existing anatomical features, as the ovary may be deformed or absent due to surgical resection, making it difficult to delineate and segment the affected areas (Liang et al., 2025). The difficulty this poses is reflected even at the human expert level; inter-rater agreement for manual ovary segmentation in endometriosis MRI has been shown to be only moderate, with pairwise Dice Similarity Coefficients as low as 0.34 between experienced raters (Liang et al., 2025). Segmentation itself is needed in order to actualise a diagnosis, as inspecting the segmented MRI allows the doctor to give a verdict on whether or not something may be present. It is pivotal, then, that this segmentation is accurate and reliable.

In recent years, segmentation in the medical field has improved considerably, and deep learning methods have demonstrated effectiveness in pelvic organ segmentation using MRI, particularly for prostate and cervical cancer (Nie et al., 2018; Huang et al., 2021; Kurata et al., 2019; Hodneland et al., 2021). However, in the realm of endometriosis, there is very little innovation, and automatic ovary segmentation methods for endometriosis patients based on MRI remain scarce (Liang et al., 2025). In 2025, Liang et al. constructed two endometriosis MRI datasets to develop a semi-automated endometriosis segmentation pipeline, with the hope of improving the diagnostic process for doctors and patients. However, the task is very difficult. As with most AI models, the task complexity correlates with the data needed to tackle it, and the training set used by Liang et al. comprised only 30 labelled subjects, far too few given the complexity of the segmentation task, and arguably the biggest shortcoming of their methodology. This data scarcity is compounded by the fact that acquiring and manually labelling endometriosis MRI scans is itself a resource-intensive process, further limiting the growth of available annotated datasets. Data scarcity can, however, be remedied in a number of ways, one of which is synthetic data generation. This project therefore proposes a new generative architecture capable of producing realistic synthetic pelvic MRI data that can augment existing training sets, with the aim of improving the performance and viability of endometriosis segmentation models.

1.2 Aims and Objectives

Aim

This project aims to develop a conditional diffusion model architecture capable of generating realistic synthetic axial pelvic MRI data and to evaluate whether augmenting limited training sets with this synthetic data improves the performance of endometriosis segmentation models.

Objectives

1.	Develop a state-of-the-art DDPM architecture for visual pelvic MRI data
2.	Analyse the quality and performance of synthetic data in downstream segmentation tasks
3.	Present an in-depth report on the results, methodology and implications of the project
4.	Create a comprehensive literature review to understand the field and the problem in depth

2. Literature Review

2.1 Literature Search Strategy

The literature search was performed along multiple academic databases, including Google Scholar, StarPlus, and arXiv. Search terms included 'pelvic MRI synthesis', 'GAN medical image generation', 'endometriosis segmentation', 'synthetic MRI augmentation', and '2D/3D MRI processing', used both individually and in combination. Results were initially parsed by title and abstract to assess relevance, with priority given to publications from 2018 onwards. Studies that had the relevance criteria were then examined in full, with their reference lists additionally reviewed to identify further pertinent works.

2.2 The Literature Review

Pelvic MRI segmentation and the data scarcity problem:

The main paper that spawned this experiment was Liang et al. This paper presented data on endometriosis and a novel segmentation pipeline. It also, in research, made me understand the general data scarcity problem for this task. Liang et al. (2025) established the UT-EndoMRI benchmark, reporting a best ovary segmentation Dice Similarity Coefficient (DSC) of just 0.290 using their RAovSeg pipeline, trained on only 30 subjects comprising 594 ovary-containing slices. Critically, inter-rater agreement among expert radiologists was only 0.48 $\pm$ 0.24, reflecting the inherent difficulty of ovary delineation, where the average ovary volume is just 12.2 cc compared to 220.3 cc for the uterus. The segmentation bottleneck is not architectural: even nnU-Net (Isensee et al., 2021), a self-configuring framework that automatically adapts to dataset characteristics, achieved only a DSC of 0.272 on this data. The gap between automated performance (0.290) and human agreement (0.48) suggests it's not an issue with the architecture, but rather with data scarcity.

Diffusion models for medical image synthesis:

Given that the datasets provided by Liang et al. are cornerstones on the endometriosis task, the question then becomes: what data augmentation techniques can be used to improve them, such as synthetic data generation? Diffusion probabilistic models have risen as the primary generative framework for medical image synthesis. Ho et al. (2020) introduced denoising diffusion probabilistic models (DDPMs), which learn to iteratively reverse a noise-diffusion process, offering better performance and stability than generative adversarial networks (GANs), which risk mode collapse in low-data regimes. Dorjsembe et al. (2024) extended this approach to 3D semantic brain MRI synthesis with Med-DDPM, conditioning generation on segmentation maps via channel concatenation to produce anatomically coherent volumes. However, Med-DDPM operates on full 3D volumes, requiring data quantities unavailable in the endometriosis domain, and its conditional generation treats all label channels equally, regardless of anatomical structure size. In contrast, Kebaili et al. (2024) demonstrated that a slice-based latent diffusion approach (SBLDM) can successfully augment tumour segmentation training data even in data-scarce regimes, validating the 2D slice-by-slice generation strategy for settings where volumetric approaches are impractical. Together, these works establish diffusion models as effective for medical synthesis but also show that existing methods have tangible setbacks, either requiring large 3D datasets or relying on conditioning mechanisms that do not prioritise small-structure fidelity.

Semantic conditioning:

One of the biggest issues I foresee with synthetic data generation for this project is the size and fidelity of the generated anatomical components. As such, semantic conditioning, the mechanism by which anatomical labels condition image generation, can be one of our biggest improvements by adding a conditioning that preserves anatomical fidelity in small sizes. Park et al. (2019) introduced spatially adaptive denormalisation (SPADE), which replaces standard normalisation layers that destroy spatial semantic information with per-pixel affine transformations learned from the input label map. This preserves pixel-level boundary correspondence, which is particularly important where structures vary dramatically in size. Li et al. (2025) directly validated this approach for medical image synthesis by combining SPADE with diffusion models to augment multi-label cardiac MRI and found that SPADE conditioning outperformed cross-attention alternatives in maintaining structural fidelity across multiple cardiac regions. However, cardiac chambers are substantially larger than ovaries, and no existing work has applied SPADE-conditioned diffusion to pelvic organ synthesis where the small-structure challenge is arguably more severe. This represents the core technical gap that the project aims to tackle.

Domain adaptation and cross-domain synthesis:

Cross-domain synthesis between imaging protocols is an established paradigm in medical imaging. Isola et al. (2017) introduced the PatchGAN discriminator, which operates on local image patches to enforce texture realism rather than global image statistics, making it effective for style transfer tasks. Huo et al. (2019) demonstrated, using SynSeg-Net, that adversarial training signals can bridge the domain gap between MRI and CT modalities, enabling segmentation without target-modality ground truth. These approaches typically apply domain adaptation as a separate post-processing step. In our project, patch gan would come into effect in phase 2, where the best-performing generative architecture would be retrained on Dataset 1, T2-weighted slices. A PatchGAN discriminator is then trained exclusively on Dataset 2, T2-weighted fat suppression slices provide a continuous style signal that transforms the different sequences between datasets turning the problem into a training feature.

The research gap:

In summary, existing diffusion-based medical image synthesis methods either use conditioning mechanisms that lack the spatial precision needed for small pelvic structures, operate in 3D, requiring data volumes unavailable for endometriosis MRI, or do not address cross-domain issues between different MRI sequences. This project addresses these limitations by combining SPADE conditioning for boundary fidelity with adversarial domain adaptation for protocol matching, within a 2D framework suitable for extreme data scarcity.

2.3 Research Question(s) / Hypotheses

To what extent can synthetic data augmentation using conditional diffusion models close the performance gap between automated ovary segmentation and inter-rater agreement in endometriosis MRI?

Augmenting the RAovSeg training set with synthetic pelvic MRI generated by a conditional diffusion model with spatially adaptive conditioning will noticeably improve ovary segmentation performance, closing the gap between the current automated baseline and inter-rater agreement.

3. Methodology Planning

3.1 Methodological Approach

This project adopts a quantitative, experimental methodology. The proposed approach centres on synthetic data augmentation to address the limited training data available for pelvic segmentation in endometriosis MRI. The proposed system is a 2D conditional denoising diffusion probabilistic model (DDPM) that takes a five-channel pelvic organ segmentation label map as input and generates a corresponding synthetic T2-weighted fat-suppressed MRI slice. Each generated image is paired with its conditioning label map, producing synthetic training data for the RAovSeg segmentation pipeline. A two-phase experimental design is employed to isolate the contributions of individual architectural components (Phase 1) and evaluate a cross-domain training strategy (Phase 2). The following will be key components of the generator and discriminator.

Diffusion model (DDPM): Diffusion models were selected over generative adversarial networks due to their superior training stability and mode coverage in low-data regimes. Where only 30 labelled training volumes are available, the risk of mode collapse associated with GAN-based approaches is prohibitive. The model learns to iteratively denoise images conditioned on anatomical label maps, producing synthetic MRI slices that correspond to specified organ boundaries (Ho et al., 2020).

SPADE conditioning: Standard normalisation layers destroy spatial semantic information, which is particularly detrimental for small structures. Given that ovary volumes average 12.2cc compared to 220.3cc for the uterus (Liang et al., 2025), preserving pixel-level boundary fidelity is critical. Spatially adaptive denormalisation (SPADE) addresses this by learning per-pixel affine parameters from the input label map (Park et al., 2019). SPADE ensures anatomical fidelity at the boundary level, while the PatchGAN discriminator (below) ensures the generated tissue textures match the target imaging protocol.

PatchGAN discriminator: An adversarial discriminator operating on local image patches is incorporated to enforce both texture realism and label-image boundary consistency. The

discriminator is trained exclusively on Dataset 2 T2w fat suppression slices, providing a domain-specific style signal that guides the generator toward the target imaging protocol. The adversarial loss is weighted as a secondary objective alongside the primary diffusion loss to maintain training stability. Together with SPADE, these components address the dual challenge of structural accuracy and domain-appropriate appearance.

3.2 Planned Data Collection Procedure

This study utilises the publicly available UT-EndoMRI dataset (Liang et al., 2025), comprising two endometriosis MRI datasets from distinct clinical institutions. No new data collection is required; all experiments use existing de-identified NIfTI volumes.

Dataset 1 comprises 51 subjects across 15 sites and 9 scanner models from two manufacturers, with T1-weighted, T1-weighted fat suppression, and T2-weighted sequences. Dataset 2 comprises 81 subjects from a single site using a single machine (Philips Ingenia 1.5T), with T2-weighted fat suppression as the primary sequence. The existing train/test split of 30 training and 8 test subjects from Dataset 2 is preserved to enable direct comparison with the published RAovSeg baseline.

Notably, Dataset 1 does not include T2-weighted fat suppression imaging. T2-weighted is the closest available sequence, sharing the same fundamental contrast weighting but lacking fat signal suppression. This issue between datasets motivates the adversarial domain adaptation component of the proposed architecture.

All volumes undergo identical preprocessing to match the published RAovSeg pipeline: intensity clipping to the 1st-99th percentile, normalisation to $[0,1]$, and resampling to 512x512 pixels at 5mm isotropic voxel spacing. The custom ovary intensity enhancement described by Liang et al. (2025) is applied as a downstream step, identically to both real and synthetic data, to ensure comparability. Five-channel binary label maps encoding background, uterus, bilateral ovaries, and endometrioma are used as conditioning inputs.

3.3 Planned Analysis Methods

Evaluation follows a two-phase ablation study designed to isolate the contribution of each architectural component. Phase 1 trains all generators on Dataset 2 under identical conditions, varying only the conditioning mechanism.

Three configurations are compared: (i) a baseline conditional DDPM with channel concatenation conditioning, (ii) DDPM with SPADE conditioning, and (iii) DDPM with SPADE conditioning and adversarial training via PatchGAN. An additional comparison is made against Med-DDPM (Dorjsembe et al., 2024), an established 3D diffusion-based synthesis method, to evaluate whether volumetric generation offers advantages despite the limited training data.

Phase 2 retrains the best-performing architecture from Phase 1, replacing the generator training data with T2w volumes from Dataset 1 while retaining Dataset 2 as the discriminator anchor, to evaluate whether cross-domain anatomical diversity improves synthetic data quality.

The primary evaluation metric is the Dice Similarity Coefficient (DSC) for ovary segmentation, measured by retraining the RAovSeg pipeline (Liang et al., 2025) on combined real and synthetic data and evaluating on the fixed 8-subject held-out test set. All experiments generate an equal number of synthetic volumes and use identical RAovSeg hyperparameters, to ensure that any performance differences are attributable to the synthetic data quality alone. Results are reported as mean $\pm$ standard deviation over three independent training runs with different random seeds.

Synthetic image quality is additionally analysed using Frechet Inception Distance, boundary fidelity DSC, and nearest-neighbour distance to detect memorisation. A titration experiment varies the ratio of synthetic to real training data to identify the optimal augmentation level.

3.4 Consideration of Ethical Issues

The UT-EndoMRI dataset was collected under IRB approval (protocol HSC-SBMI-22-0184) with appropriate patient consent. All data is de-identified and publicly available in NIfTI format. This study involves no new data collection, no patient contact, and no processing of identifiable information.

3.5 Project Timetable / Schedule and Milestones

The project follows a seven-week experimental schedule:

Week 0: Finalise literature review.

Week 1: Data preparation and baseline reproduction.

Week 2: Experiment 1a -- DDPM with channel concatenation conditioning.

Week 3: Experiment 1b -- DDPM with SPADE conditioning.

Week 4: Experiment 1c -- DDPM with SPADE + PatchGAN, plus Med-DDPM comparison.

Week 5: Phase 1 evaluation and Phase 2 cross-domain training.

Week 6: Titration experiments, quality metrics, and results compilation.

This schedule includes built-in de-risking: Phase 1 validates the core approach before Phase 2 introduces cross-domain complexity. It is noteworthy that the timeline includes a $\pm$ 4-week buffer, as training and generation times are currently unknown.

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